

Kishalay Das

Research Interests

I am a **Postdoctoral Associate** at **Yale University** (Krishnaswamy Lab). My research lies at the intersection of **Graph Representation Learning**, **Geometric Deep Learning**, and **Generative Modeling**, with applications in **AI for Science** and **AI for Medicine** — particularly scalable material generation and property prediction.

Academic Appointments

- Apr 2026 – present **Postdoctoral Associate**, *Yale University*, New Haven, CT, USA.
Krishnaswamy Lab · Supervisor: Prof. Smita Krishnaswamy
- Jul 2020 – Dec 2021 **Research Assistant**, *IIT Kharagpur*, Kharagpur, India.
CNeRG Group

Education

- 2022 – 2026 **Ph.D. in Computer Science**, *Indian Institute of Technology (IIT) Kharagpur*, India,
GPA: 9.5 / 10.
Thesis: AI for Scalable Material Generation and Property Prediction
Advisors: Prof. Niloy Ganguly & Prof. Pawan Goyal
- 2018 – 2020 **M.Tech. in Computer Science**, *Indian Institute of Science (IISc)*, Bangalore, India,
GPA: 8.8 / 10.
Thesis: Graph Neural Network on Hypergraphs by Learning Line Graph Expansion
Advisor: Prof. M. Narasimha Murty
- 2008 – 2012 **B.Tech. in Computer Science**, *West Bengal University of Technology*, Kolkata, India,
GPA: 8.37 / 10.

Publications

* denotes equal contribution.

- S. Mukherjee*, K. Das*, P. Basuchowdhuri, P. Goyal, N. Ganguly. **Model Agnostic Graph Prompt Learning for Crystal Property Prediction**. *UAI 2026*. [Paper]
- S. Mukherjee*, K. Das*, P. Basuchowdhuri, P. Goyal, N. Ganguly. **Latent Diffusion Pretraining for Crystal Property Prediction**. *ICML 2026; AI4MAT Workshop @ ICLR 2026*. [Paper]
- S. Khastagir*, K. Das*, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **LLM Meets Diffusion: A Hybrid Framework for Crystal Material Generation**. *NeurIPS 2025*. [Paper] [Code]
- K. Das, S. Khastagir, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **CrysLDM: Latent Diffusion Model for Crystal Material Generation**. *AI4MAT Workshop @ ICLR 2025*. [Paper]
- K. Das, S. Khastagir, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **Periodic Materials Generation using Text-Guided Joint Diffusion Model**. *ICLR 2025*. [Paper] [Code]
- K. Das, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **CrysMMNet: Multimodal Representation for Crystal Property Prediction**. *UAI 2023*. [Paper] [Code]
- K. Das, B. Samanta, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **CrysGNN: Distilling Pre-trained Knowledge to Enhance Property Prediction for Crystalline Materials**. *AAAI 2023; ML4Materials Workshop @ ICLR 2023*. [Paper] [Code]
- K. Das, B. Samanta, P. Goyal, S.-C. Lee, S. Bhattacharjee, N. Ganguly. **CrysXPP: An Explainable Property Predictor for Crystalline Materials**. *npj Computational Materials*, 2022. [Paper] [Code]
- S. Bandyopadhyay, K. Das, M. N. Murty. **Hypergraph Attention Isomorphism Network by Learning Line Graph Expansion**. *IEEE Big Data 2020*. [Paper] [Code]
- S. Bandyopadhyay, K. Das, M. N. Murty. **Line Hypergraph Convolution Network: Applying Graph Con-**

volution for Hypergraphs. *arXiv*, 2020. [\[Paper\]](#)

Industry Experience

May 2017 – Jul 2018 **Scientist/Engineer-SC**, *Indian Space Research Organisation (ISRO)*, Bangalore, India.
Mar 2013 – Apr 2015 **System Engineer**, *Tata Consultancy Services (TCS)*, Kolkata, India.
Jul 2012 – Feb 2013 **Associate Software Developer**, *Accenture Pvt. Ltd.*, Hyderabad, India.

Academic Service

Program Committee AAAI 2026; IJCAI 2025; AAAI 2025
Reviewer NeurIPS 2026; ICLR 2026; NeurIPS 2025; ICLR 2025; LOG 2025; KDD 2024; AAAI 2024; LOG 2024

Honors & Awards

- **ICLR Travel Grant**, ICLR 2025
- **ACM/IARCS Travel Grant**, ICLR 2025
- **Top Reviewer**, Learning on Graphs (LOG) 2024
- **ACM/IARCS Travel Grant**, UAI 2023
- **Google Travel Grant**, UAI 2023
- **Prime Minister's Research Fellowship (PMRF)**, 2022
- **Best Grade (A+)** on M.Tech. Thesis, IISc Bangalore, 2020
- **Outstanding** annual performance rating, ISRO, 2018
- **GATE-CS All India Rank 37** (out of 96,878 candidates), 2017
- **ISRO Entrance Exam All India Rank 38** (out of ~35,000 candidates), 2017

References

Prof. Niloy Ganguly
Professor, IIT Kharagpur, India

Prof. Pawan Goyal
Associate Professor, IIT Kharagpur, India

Dr. Satadeep Bhattacharjee
R&D Team Leader, IKST, South Korea

Prof. M. Narasimha Murty
Honorary Professor, IISc Bangalore, India